

Case Study Extraction Method v2: Concrete Ranking-Shift Example from Platform Data

Platform X Research

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1 Why we need this

The paper’s central claim—frozen opponent pools mis-rank current policies—rests on aggregate statistics (Spearman ρ , inversion rates). A reviewer will ask: “*Show me one policy whose rank actually changed because the pool moved.*” This document gives the exact procedure to produce that example from data already on disk.

No new matches need to be run. All pairwise win-rate matrices are in A.REQ3.

2 Recommended environment: Snake 3v3 (env_id 6)

This is the **first choice** because:

- It has the richest longitudinal data: quarterly payoff matrices from 2021Q1 (5 strategies) to 2026Q2 (118 strategies), plus a final snapshot of 121.
- The controlled old-vs-recent-pool experiment (Table ?? in the paper) already used this environment and found $\Delta\rho = 0.08$. We already know the effect exists; we just need to name one policy.
- The target policies, old pool, recent pool, and reference pool are **already fixed** in `cohorts.json` (`static_living_results`). No new cohort selection needed.

2.1 Exact data to use

From `static_living_results_20260928/cohorts.json`, scenario `all_history_first10`, env 6:

Set	Size	Submission IDs
Targets T	15	11381, 11391, 11398, 11408, 11430, 11436, 11438, 11439, 11444, 11446, 11448, 11449, 11451, 11452, 11453
Old pool $\mathcal{P}_{old}$	20	931, 948, 985, 1016, 1116, 1160, 1163, 1226, 1333, 1364, 1401, 1413, 1470, 1474, 1511, 1555, 1743, 1757, 1780, 2222
Recent pool $\mathcal{P}_{rec}$	20	10104, 10264, 10285, 10747, 10755, 10916, 10920, 10964, 10970, 11094, 11100, 11179, 11201, 11205, 11273, 11285, 11286, 11324, 11352, 11362
Reference pool $\mathcal{P}_{ref}$	81	(in <code>cohorts.json</code>)

Key dates: targets submitted 2022-01-02; old pool from 2021-07 to 2021-08; recent pool from 2021-12 to 2022-01; reference from 2021-08 to 2021-12. The old–recent gap is **5 months**.

2.2 What to compute

For each of the 15 target policies $t \in T$:

1. Compute $s_{old}(t)$ = average win rate of t against $\mathcal{P}_{old}$ (20 opponents).
2. Compute $s_{rec}(t)$ = average win rate of t against $\mathcal{P}_{rec}$ (20 opponents).
3. Compute $s_{ref}(t)$ = average win rate of t against $\mathcal{P}_{ref}$ (81 opponents).
4. Rank the 15 targets by each score: R_{old} , R_{rec} , R_{ref} .

The aggregate result is already known: $\text{Spearman}(R_{old}, R_{ref}) = 0.856$; $\text{Spearman}(R_{rec}, R_{ref}) = 0.933$. We need the **per-target** rank table.

2.3 Pick the hero policy

From the per-target table, select the target with the largest absolute rank displacement:

$$t^* = \arg \max_{t \in T} |\text{rank}_{old}(t) - \text{rank}_{rec}(t)|.$$

Expected finding: some target is ranked e.g. #3/15 against the old pool but #10/15 against the recent pool, while its reference rank is #8. The story: *“Against 5-month-old opponents, this policy looked like a top-3 performer. Against equally fresh opponents, it was mid-pack. The independent reference pool confirms the recent ranking.”*

If no single target has a dramatic shift (all within ± 2 ranks), use the **median** rank displacement across all 15 targets and report it as a distribution, not a hero story.

2.4 Diagnose which recent opponents caused the shift

For t^* , list its win rate against each of the 20 recent-pool opponents. Identify the 2–3 recent opponents that beat t^* (win rate < 0.4) and check: do these same opponents exist in the old pool? (They shouldn’t—recent pool is from 5 months later.) This shows the drop is caused by new entrants, not by t^* weakening.

3 Fallback: Olympics-integrated (env_id 88)

If Snake 3v3’s rank shifts are too small (because τ is low), use env 88 instead:

- 38 strategies in the latest snapshot, growing from 10 in 2022Q2.
- Quarterly matrices available for 2022Q2 through 2026Q2.
- This environment also appeared in the controlled experiment (old $\rho = 0.941$, recent $\rho = 0.986$).
- The static_living cohorts already define 6 targets, 5 old opponents, 5 recent opponents, 23 reference opponents.

Same procedure: compute per-target ranks, pick the one with biggest shift. The smaller pool (6 targets) means each rank position counts more.

4 What NOT to use

- **GRF 11v11 (env 34)**: only 25 strategies in the latest snapshot; the controlled experiment showed both old and recent pools agree ($\rho > 0.95$) because the old–recent gap is only 12 months. Not enough variance.
- **Football 5v5 (env 71)**: only 26 strategies.
- **Competition winners from E2**: their submission IDs (e.g., 25677, 25687) are **not** in the A_REQ3 daily-submission matrices. The matrices only contain daily submissions, not competition submissions. Reconstructing their head-to-head results would require querying the full match log, which is outside this data package.

5 Output

The collaborator should produce:

1. A CSV with 15 rows (or 6 for env 88): `submission_id`, `rank_old`, `rank_rec`, `rank_ref`, `win_rate_old`, `win_rate_rec`, `win_rate_ref`.
2. One bar chart: horizontal bars showing `rank_old`, `rank_rec`, `rank_ref` for t^* (the hero).
3. A 2-sentence annotation: which new opponents beat t^* .

Drop into paper §4.1 after Table ??.

6 If the hero is boring

If no target shows a rank shift >3 positions, do **not** force a hero story. Instead:

1. Report the full distribution of rank shifts (median, IQR) across all 15 targets.
2. State honestly: *“In this low- τ environment, individual rank shifts are modest (median ± 2 positions), but systematic agreement with the reference pool degrades from $\rho = 0.933$ to 0.856.”*
3. This is still a valid result: it shows that even where individual shifts are small, the aggregate ranking degrades. The high- τ environments (GRF, Olympics) would show larger per-policy shifts, but their sample sizes are too small for a controlled study.